Maths IA-1 — THE ONLY SHEET (one night, one source, final)

> **CLOSE EVERYTHING ELSE.** This file is your entire night and your entire exam prep: schedule · 27 quick traps · 8 hard questions · 4 applications · playbook · formula card. Every question now has: **TRIGGER** (how the paper signals this type) · **STEPS** (full solution) · **REMEMBER** (the line that scores the mark).

NIGHT SCHEDULE — follow the clock, no decisions needed

Block	Time	Do
I	now – +2.5h	TIER 1: 27 quick traps (read TRIGGER → cover STEPS → solve → check)
—	+2.5h – +2.75h	break: walk, water, food
II	+2.75h – +6h	TIER 2 (8 hard) + TIER 3 (4 applications), full steps
III	+6h – +6.5h	re-solve EVERY wrong one with the answer covered
IV	+6.5h – +7.5h	memorise FORMULA CARD + PLAYBOOK
V	+7.5h – +8.5h	timed re-run Tier 1, 25 min, zero errors
VI	+8.5h – +9h	timed re-run 5 best Tier 2, 30 min
VII	final 30 min	formula card glance, get ready, leave

TIER 1 — 27 quick-mark traps (the safe marks in ANY format)

1. If $a < b$, which need NOT be true? (a) $a-3 < b-3$ (b) $-4a > -4b$ (c) $a^2 < b^2$ (d) $3a < 3b$

- **TRIGGER:** wording "need NOT"/"always" → test with $a = -5$, $b = 1$
- **STEPS:** (a) $-8 < -2$ ✓ · (b) $20 > -4$ ✓ (flip works) · (c) $25 < 1$ ✗ FALSE → answer **(c)** · (d) $-15 < 3$ ✓
- **REMEMBER:** × negative flips, × positive keeps, **squaring breaks order when signs differ**

2. $\log_2(x-1) = 3 \rightarrow x = ?$

- **TRIGGER:** log of an expression = number → drop the log first
- **STEPS:** $x - 1 = 2^3 = 8 \rightarrow x =$ **9**
- **REMEMBER:** undo log BEFORE touching the +1; **8 is the bait answer**

3. $x^2 + (k-3)x + k = 0$ has equal roots → sum of possible k = ? product = ?

- **TRIGGER:** "equal roots" / "touches the axis once"
- **STEPS:** $D = (k-3)^2 - 4k = 0 \rightarrow k^2 - 6k + 9 - 4k = k^2 - 10k + 9 = 0 \rightarrow k = 1, 9 \rightarrow$ sum = **10**, product = **9**
- **REMEMBER:** D = 0 engine; **underline sum vs product BEFORE solving**

4. Solution of $(x-3)/(x+1) < 0 \rightarrow$

- **TRIGGER:** rational < 0 → numerator and denominator opposite signs → between the roots
- **STEPS:** roots −1 and 3 → test x = 0: (−3)/(1) = −3 < 0 ✓ inside → **−1 < x < 3**, x ≠ −1
- **REMEMBER:** denominator point NEVER included; no flip ever happens here

5. Digits of 2^{100} (log 2 = 0.3010) →

- **TRIGGER:** "how many digits"
- **STEPS:** $\log(2^{100}) = 30.10 \rightarrow \lfloor 30.10 \rfloor + 1 =$ **31**
- **REMEMBER:** digits = $\lfloor \log N \rfloor + 1$ — the **+1** is the whole question

6. Last two digits of 7^{200} →

- **TRIGGER:** "last two digits" → mod 100
- **STEPS:** $7^4 = 2401 \equiv 01 \pmod{100} \rightarrow 7^{200} = (7^4)^{50} \equiv 1^{50} =$ **01**
- **REMEMBER:** memorise hero $7^4 \equiv 01$ — one line, full marks

7. gcd(a,b) = 12, lcm(a,b) = 120 → ordered pairs (a,b) = ?

- **TRIGGER:** gcd AND lcm both given → pull the gcd out
- **STEPS:** a = 12x, b = 12y with xy = 10 and gcd(x,y) = 1 → (1,10)(2,5)(5,2)(10,1) → **4**
- **REMEMBER:** divide both by the gcd first, then coprime-factor the rest

8. "All multiples of 4 are even. 12 is even. So 12 is a multiple of 4." Why invalid?

- **TRIGGER:** "reasoning incorrect/flawed" → find counterexample
- **STEPS:** 6 is even but not a multiple of 4 → the conclusion doesn't follow → **(b)**
- **REMEMBER:** going backwards from a property needs proof, not vibes — cite the counterexample

9. |x| = −x exactly when →

- **TRIGGER:** absolute value equals its negative
- **STEPS:** x = 0: 0 = 0 ✓; x < 0: |x| = −x ✓; x > 0: x = −x ✗ → **x ≤ 0**
- **REMEMBER:** **0 is included** — trap answer is x < 0

10. $x + 1/x = 4 \rightarrow x^3 + 1/x^3 = ?$

- **TRIGGER:** x + 1/x given, higher power asked
- **STEPS:** $(x+1/x)^3 = x^3 + 1/x^3 + 3(x+1/x) \rightarrow 64 =$ ans + 12 → **52**
- **REMEMBER:** cube formula subtracts **3×(the given)**, not 3

11. $\log_5 125 + \log_2 32 - \log_{10} 10000 = ?$

- **TRIGGER:** logs of exact powers → convert to exponents
- **STEPS:** $5^3=125 \rightarrow 3 \cdot 2^5=32 \rightarrow 5 \cdot 10^4=10000 \rightarrow 4 \rightarrow 3 + 5 - 4 =$ **4**
- **REMEMBER:** rewrite each as "to what power" — arithmetic then trivial

12. Remainder of $10^{10} \div 7 = ?$

- **TRIGGER:** big power remainder → reduce base, find cycle
- **STEPS:** $10 \equiv 3 \pmod 7$; $3^6 = 729 \equiv 1$; $10 = 6\times1 + 4 \rightarrow 3^4 = 81 = 7\times11 + 4 \rightarrow$ **4**
- **REMEMBER:** reduce base first, then exponent mod cycle length (6 here)

13. Divisors of 360 = ?

- **TRIGGER:** "number of divisors" → prime-factorise
- **STEPS:** $360 = 2^3 \cdot 3^2 \cdot 5 \rightarrow (3+1)(2+1)(1+1) =$ **24**
- **REMEMBER:** τ = product of (exponent + 1) — always +1

14. (x − 2) divides $x^3 - kx^2 + 10x - 8$ exactly → k = ?

- **TRIGGER:** "(x – a) divides/exactly" → factor theorem
- **STEPS:** f(2) = 8 – 4k + 20 – 8 = 0 → 20 – 4k = 0 → k = **5**
- **REMEMBER:** plug the ROOT (sign flips: x–2=0 → x=+2), set whole thing = 0

15. 4^(x+1) = 8^x → x = ?

- **TRIGGER:** power = power → make bases SAME
- **STEPS:** 2^(2x+2) = 2^(3x) → 2x + 2 = 3x → x = **2**
- **REMEMBER:** bases equal ⟹ exponents equal — never subtract bases

16. Product of roots of $3x^2 – 7x + 2 = 0$ →

- **TRIGGER:** roots asked WITHOUT solving → Vieta's formulas
- **STEPS:** product = c/a = **2/3** (sum would be 7/3)
- **REMEMBER:** product = c/a, sum = –b/a — sign is the trap

17. MATCH: a) x+1/x=3 → x^2+1/x^2 · b) lcm(12,18) · c) 100 mod 7 · d) $\log_9 27$ · e) divisors of 60 — vs 7 / 36 / 2 / 3/2 / 12

- **TRIGGER:** matching → solve each pair INDEPENDENTLY, never chain
- **STEPS:** a) 9 – 2 = **7** · b) **36** · c) 100 = 14×7 + 2 → **2** · d) 9^{3/2}=27 → **3/2** · e) $2^2 \cdot 3 \cdot 5$ → 2·2·2 = **12**
- **REMEMBER:** pair-by-pair, lock the confident 3 first

18. Complete the square: $x^2 – 8x + 5$ →

- **TRIGGER:** "write in the form $(x + p)^2 + q$"
- **STEPS:** half of –8 = –4 → $(x – 4)^2 – 16 + 5$ = **$(x – 4)^2 – 11$**
- **REMEMBER:** halve, square, SUBTRACT what you added

19. gcd(1071, 462) with Euclidean steps →

- **TRIGGER:** "use Euclidean algorithm" / big numbers
- **STEPS:** 1071 = 2×462 + 147 → 462 = 3×147 + 21 → 147 = 7×21 + 0 → **21**
- **REMEMBER:** write every division line — the steps are the marks, answer alone ≈ half credit

20. Integers 1–100 divisible by 6 OR by 8 →

- **TRIGGER:** "or" in counting → inclusion–exclusion
- **STEPS:** ⌊100/6⌋ = 16, ⌊100/8⌋ = 12, ⌊100/24⌋ = 4 → 16 + 12 – 4 = **24**
- **REMEMBER:** |A| + |B| – |A∩B| — subtracting the double-count IS the question

21. Find k so $x^2 + 2x + k > 0$ for EVERY real x →

- **TRIGGER:** "for all real x" / "always positive"
- **STEPS:** need D < 0 and open-up: 4 – 4k < 0 → **k > 1**
- **REMEMBER:** strict — k = 1 fails (x = –1 gives exactly 0, not > 0)

22. Factorise fully: $x^4 – 16$ →

- **TRIGGER:** "fully" → apply difference of squares until it stops
- **STEPS:** $x^4 – 16 = (x^2)^2 – 4^2 = (x^2–4)(x^2+4)$ = **$(x–2)(x+2)(x^2+4)$**
- **REMEMBER:** $x^2 + 4$ never factors over reals — stop there

23. $2^3 \cdot 2^5 \div 2^4$ = →

- **TRIGGER:** same base, × ÷ → add/subtract exponents
- **STEPS:** 3 + 5 – 4 = 4 → 2^4 = **16**
- **REMEMBER:** never multiply the bases — only the powers move

24. Express log 45 using log 3 and log 5 →

- **TRIGGER:** "in terms of" → factorise the number
- **STEPS:** $45 = 9 \times 5 = 3^2 \cdot 5$ → $2 \log 3 + \log 5$
- **REMEMBER:** split the NUMBER, not the log

25. If $2^x = 7$ then x = →

- **TRIGGER:** exponential given, log form asked
- **STEPS:** $a^b = c \iff b = \log_a c$ → x = $\mathbf{\log_2 7}$ (or log 7 / log 2)
- **REMEMBER:** base of log = base of power, always

26. gcd(a,b) = 8, lcm(a,b) = 96, a = 24 → b = ?

- **TRIGGER:** three of the four gcd/lcm quantities → product rule
- **STEPS:** a·b = gcd·lcm → $24 \cdot b = 8 \times 96 = 768$ → b = **32**
- **REMEMBER:** gcd × lcm = a × b — the one law that binds them

27. Buses leave every 15 min and 20 min, both at 8:00 → next together at?

- **TRIGGER:** "next time together" → LCM of the intervals
- **STEPS:** lcm(15, 20) = 60 → 8:00 + 60 min = **9:00**
- **REMEMBER:** together = LCM, ever-decreasing gaps = HCF — never mix them

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TIER 2 — 8 hard questions (the heavy-marks engine; exam needs 2–3 perfect)

Full-mark rule: GIVEN → FORMULA → STEPS → ANSWER → one sentence of meaning.

H1. Last two digits of 31^{31}.

- **TRIGGER:** last two digits of a NON-8-or-7 base → binomial split
- **STEPS:** 31 = 30 + 1 → $(30+1)^{31} = 1 + 31 \times 30 +$ terms with $30^2 = 900 \equiv 0 \pmod{100}$ → 1 + 930 = 931 → **31**
- **REMEMBER:** split into (tens + ones); everything with 30^2 dies mod 100

H2. Prove $n^4 - n^2$ divisible by 12 for every integer n.

- **TRIGGER:** "prove divisible by" → factorise into consecutive products
- **STEPS:** $n^2(n-1)(n+1)$: $(n-1)n(n+1)$ gives a factor 3; n even → n^2 gives 4, n odd → two even neighbours give 4 → divisible by 3 AND 4, coprime → **12** ∎
- **REMEMBER:** split the divisor (12 = 3 × 4) and prove each part separately

H3. Greatest number dividing 552, 967, 1032 leaving the SAME remainder?

- **TRIGGER:** "same remainder" → gcd of DIFFERENCES
- **STEPS:** differences 415, 65, 480 → Euclid(415,65): 415 = 6×65+25 → 65 = 2×25+15 → 25 = 15+10 → 15 = 10+5 → **5**; check 552=110×5+2, 967=193×5+2, 1032=206×5+2 all leave 2 ✓
- **REMEMBER:** answer is gcd(differences) — **always verify remainders match** (that check is one mark)

H4. Same distance upstream in 4 h as downstream in 2 h; stream 3 km/h. Boat speed & distance?

- **TRIGGER:** two trips, same distance → set the products equal
- **STEPS:** 4(b − 3) = 2(b + 3) → 4b − 12 = 2b + 6 → 2b = 18 → b = **9 km/h**; distance = 4 × 6 = **24 km**; check 2 × 12 = 24 ✓

- **REMEMBER:** with-stream = +, against = –; **write the check line**

H5. Solve log x + log(x + 3) = 1.

- **TRIGGER:** logs added → combine → solve quadratic → DOMAIN CHECK
- **STEPS:** log[x(x+3)] = 1 → $x^2 + 3x = 10$ → $x^2 + 3x - 10 = 0$ → (x+5)(x–2) = 0 → x = –5 or 2; need x > 0 → **x = 2**
- **REMEMBER:** rejection line scores a mark — "x = –5 rejected, log undefined for x < 0"

H6. Solve 5x ≡ 3 (mod 8).

- **TRIGGER:** linear congruence → need the inverse
- **STEPS:** 5×5 = 25 ≡ 1 (mod 8) → multiply both sides by 5: x ≡ 15 ≡ 7 → **x ≡ 7 (mod 8)**
- **REMEMBER:** find what multiplies to ≡ 1 — try small numbers: 1,3,5,7 (odd residues are self-inverse here)

H7. Four bells toll together at noon, then every 6, 8, 12, 15 s. Together how many times in 1 hour including start?

- **TRIGGER:** "together" + intervals → LCM + count including t = 0
- **STEPS:** lcm(6,8,12,15) = 120 s → 3600 ÷ 120 = 30 → +1 for the start = **31**
- **REMEMBER:** decide inclusion explicitly ("including the start") — +1 or not is the mark

H8. Sum of a number and its reciprocal is 10/3. Find it.

- **TRIGGER:** "sum of a number and its reciprocal" → clear fractions
- **STEPS:** $x + 1/x = 10/3$ → $3x^2 - 10x + 3 = 0$ → (3x–1)(x–3) = 0 → x = 3 or 1/3 → **both**
- **REMEMBER:** word problems can have TWO answers — give both, don't pick

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TIER 3 — 4 application questions (central teams LOVE worded problems)

T1. Garden is 5 m longer than wide; area 36 m². Dimensions? Widened to 40 m² → width?

- **TRIGGER:** area word problem → quadratic
- **STEPS:** w(w+5) = 36 → $w^2 + 5w - 36 = 0$ → (w+9)(w–4) = 0 → w = **4 m, l = 9 m** (reject –9); then $w^2 + 5w - 40 = 0$ → w = (–5 + √185)/2 ≈ **4.3 m**
- **REMEMBER:** reject the negative length in a sentence, every time

T2. IQ = MA/CA × 100. Age 15, 90 ≤ IQ ≤ 150 → mental-age range?

- **TRIGGER:** chained inequality with a formula
- **STEPS:** MA = IQ × 15/100 → 90→13.5, 150→22.5 → **13.5 ≤ MA ≤ 22.5 years**
- **REMEMBER:** substitute the ENDPOINTS only — inequalities map to endpoints

T3. $h(t) = -5t^2 + 20t + 25$: (i) zeros (ii) max height no calculus (iii) h(6) = –35 — "35 m underground"?

- **TRIGGER:** physics worded quadratic → always interpret in context
- **STEPS:** (i) –5(t–5)(t+1) → t = **5 s** (reject –1: time ≥ 0) · (ii) t = –b/2a = 2 → h(2) = 45 → **45 m** · (iii) **NO** — formula valid only during flight; t = 6 is past landing, negative height is model extrapolation
- **REMEMBER:** interpretation questions need a full SENTENCE — the "why" is the mark

T4. $pH = -\log_{10}[H^+]$. A: 10^{-3}, B: $10^{-4.5}$. Which more acidic, by how much?

- **TRIGGER:** log in a real formula → compute both, then COMPARE
- **STEPS:** pH A = 3, pH B = 4.5 → lower pH = more acidic → **A by 1.5 units**
- **REMEMBER:** state direction ("lower pH is more acidic") — comparison needs reasoning, not just numbers

PLAYBOOK — extracting 70% from an unpredictable, unknown-format, unknown-marks paper

Universal order:

1. **First 5 min:** read the WHOLE paper, note today's format, count what exists. No solving.
2. **Safe pass (~⅓ of time):** every question answerable in ≤ 90 s — in ANY order. This is your floor.
3. **Matching (if present):** pair independently, lock 3+ first.
4. **Heavy pass (~half):** pick 2–3 whose chapter you recognise → full steps. Then fight the rest.
5. **Final ~10%:** every blank gets SOMETHING; check signs, rejections, units.

70% checkpoint (mid-exam): safe pass locked + two heavy ones fully written = on track. If behind → heavy ones next, never sideways.

Twist protocol (confusing wording): read twice → **underline what's actually asked** (sum? product? "need NOT"?) → name the chapter → write givens + the one formula → solve. Still lost? Write formula + substitution anyway — partial credit is real credit.

Format-switch rules: MCQ with no stated negative marking → never blank, eliminate 2 first · "explain/justify" → full sentence naming the reason (domain, flip, sign, rejection, converse) · two candidate answers → plug into the ORIGINAL statement · freak format → it's chapters 1–6 underneath: identify → formula card.

3-minute rule: not cracked in 3 min → abandon, flag, return at minute 75.

FORMULA CARD — memorise cold in Block IV

Ch1: $(x+1/x)^2 - 2$ · $(x+1/x)^3 - 3(x+1/x)$ · $(a+b)^2 - 2ab$ · $(a+b)^2 - (a-b)^2 = 4ab$ · complete square: halve, square, subtract · digits = $\lfloor \log \rfloor + 1$

Ch2: $D = b^2 - 4ac$ (touch $\iff 0$, none $\iff <0$, distinct $\iff >0$) · sum = $-b/a$ · product = c/a · vertex = $-b/2a$ · both roots + $\iff$ sum>0, product>0, $D \geq 0$

Ch3: flip ONLY on ×÷ by negative · $|x| \leq a \iff -a \leq x \leq a$ · quadratic < 0 between roots · rational: sign chart + exclude denominator · "for all x" $\iff D<0$ & $a>0$

Ch4: + logs → multiply, − → divide · $\log_a a = 1$ · $a^b = N \iff b = \log_a N$ · chain collapse · digits formula · **domain first, reject negatives/zero**

Ch5: gcd × lcm = a·b · Euclid: divide until remainder 0 · same remainder $\Longrightarrow$ gcd of differences · $\tau = \Pi(e_i+1)$ · "or" count = A + B − both · pair form = gcd·x, gcd·y (coprime)

Ch6: reduce base mod n → reduce exponent mod cycle · last-digit cycles length 4 · heroes: $7^4 \equiv 01$, $3^{20} \equiv 01$ · inverse = multiply to $\equiv 1$ · $ax \equiv b$: divide by gcd first · consecutive-products proofs

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This is the whole plan. Tiers 1-2-3 in order, then playbook, then formula card. Nothing else exists tonight.